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ME2/M21

4.1.1. Pandas - series creation and manipulation

03:12

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the groupby and mean() operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

```
# import pandas as pd
```

```
# # Take inputs from the user to create a list of numbers
```

```
# numbers = list(map(int, input().split()))
```

```
# grouped =
```

```
# # Display the mean of even and odd numbers with labels
```

```
# grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
```

```

# print("Mean of even and odd numbers:")

# print(grouped)

import pandas as pd

# Take inputs from the user to create a list of numbers
numbers = list(map(int, input().split()))

# Create a Pandas series from the list of numbers
s = pd.Series(numbers)

# Grouping by even and odd numbers and calculating the mean
grouped = s.groupby(s % 2 == 0).mean()

# Display the mean of even and odd numbers with labels grouped.index =
['Even' if is_even else 'Odd' for is_even in grouped.index]

print("Mean of even and odd numbers:")
print(grouped)

```



✓ Test case 2 13 ms Debug 	
Expected output	Actual output
4 4 4 6 6 2 2 2 10 12 14 16	4 4 4 6 6 2 2 2 10 12 14 16
Mean of even and odd numbers:	Mean of even and odd numbers:
Even ... 6.833333	Even ... 6.833333
dtype: float64	dtype: float64

✓ Test case 3 13 ms Debug 	
Expected output	Actual output
1 5 7 9 11 13 15 17 111 21 25	1 5 7 9 11 13 15 17 111 21 25
Mean of even and odd numbers:	Mean of even and odd numbers:
Odd ... 21.363636	Odd ... 21.363636
dtype: float64	dtype: float64

4.1.2. Dictionary to dataframe

07:26

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age

value from the user.

- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column Gender with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the Age column.
- Display the DataFrame after deleting the column. import pandas as pd

Provided dictionary of lists

```
data = {  
  
    'Name': ['Alice', 'Bob', 'Charlie'],  
    'Age': [25, 30, 35],  
}
```

Convert the dictionary to a DataFrame

```
df = pd.DataFrame(data)
```

Display the original DataFrame

```
print("Original DataFrame:")
```

```
print(df)
```

```
# Adding a new row new_name =
```

```
input("New name: ") new_age =
```

```
int(input("New age: ")) df.loc[len(df)] =
```

```
[new_name, new_age]
```

```
# Display the DataFrame after adding a new row
```

```
print("After adding a row:\n",df)
```

```
# Modifying a row row_to_modify = int(input("Index
```

```
of row to modify: ")) new_age_value =
```

```
int(input("New age: ")) df.at[row_to_modify, "Age"] =
```

```
new_age_value
```

```
# Display the DataFrame after modifying a row
```

```
print("After modifying a row:") print(df)
```

```
# Deleting a row row_to_delete = int(input("Index of row to
```

```
delete: ")) df =
```

```
df.drop(index=row_to_delete).reset_index(drop=True)
```

```
# Display the DataFrame after deleting a row
```

```
print("After deleting a row:") print(df)
```

```
# Adding a new column genders_input = input("Enter
```

```
genders separated by space: ") genders =
```

```
genders_input.split()
```

```
# Ensure correct length if
```

```
len(genders) == len(df):
```

```
df["Gender"] = genders
else:
    print("Error: Number of genders must match rows!")
df["Gender"] = ["Unknown"] * len(df)
```

```
# Display the DataFrame after adding a new column
print("After adding a new column:")

print(df)
```

```
# Modifying a column df["Name"] =
df["Name"].str.upper()
```

```
# Display the DataFrame after modifying a column
print("After modifying a column:") print(df)
```

```
# Deleting a column df =
df.drop(columns=["Age"])
```

```
# Display the DataFrame after deleting a column
print("After deleting a column:")
print(df)
```

Explorer

Average time
0.043 s
43.00 ms

Maximum time
0.049 s
49.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ 1 out of 1 hidden test case(s) passed

Debugger

After modifying a row:

	Name	Age
0	Alice	27
1	Bob	30
2	Charlie	35
3	Susan	29

Index of row to delete: 3

After deleting a row:

	Name	Age
0	Alice	27
1	Bob	30
2	Charlie	35

Enter genders separated by space: F M M

After adding a new column:

	Name	Age	Gender
0	Alice	27	F
1	Bob	30	M
2	Charlie	35	M

After modifying a column:

	Name	Age	Gender
0	ALICE	27	F
1	BOB	30	M
2	CHARLIE	35	M

After deleting a column:

	Name	Gender
0	ALICE	F
1	BOB	M
2	CHARLIE	M

Terminal

Test cases

4.1.3. Student Information

01:49

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:

Refer to the displayed test cases for better understanding.

```
import pandas as pd
```

```
# Read the text file into a DataFrame file = input() data = pd.read_csv(file,
sep="\s+", header=None, names=["Name", "Age", "Grade"])
```

```
# Display the first five rows
```

```
print("First five rows:")
```

```
print(data.head())
```

```
# Calculate the average age (2 decimal places)
```

```
avg_age = round(data["Age"].mean(), 2)
```

```
print("Average age:", avg_age)
```

```
# Filter students with grade up to 'B' (A and B)
```

```
filtered = data[data["Grade"].isin(['A', 'B'])]
```

```
print("Students with a grade up to B")
```

```
print(filtered)
```


Average time: **0.028 s** 27.50 ms Maximum time: **0.040 s** 40.00 ms

1 out of 1 shown test case(s) passed
1 out of 1 hidden test case(s) passed

Test case 1 40 ms Debug

Expected output	Actual output
studentdata.txt	studentdata.txt
First five rows:	First five rows:
0 John 25 A	0 John 25 A
1 Alin 22 B	1 Alin 22 B
2 Emma 24 A	2 Emma 24 A
3 Mike 23 C	3 Mike 23 C
4 Sony 21 B	4 Sony 21 B
Average age: 23.29	Average age: 23.29
Students with a grade up to B	Students with a grade up to B
0 John 25 A	0 John 25 A
1 Alin 22 B	1 Alin 22 B
2 Emma 24 A	2 Emma 24 A
4 Sony 21 B	4 Sony 21 B
5 Amia 26 A	5 Amia 26 A

Terminal Test cases

< Prev Reset Submit Next >

4.1.3. Student Information

01:49

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:

Refer to the displayed test cases for better understanding.

```

import pandas as pd

# Prompt the user for the file name
file_name = input()

# Load the data df =
pd.read_csv(file_name) # Prepare
data for calculation df['Date'] =
pd.to_datetime(df['Date']) df['Month'] =
df['Date'].dt.to_period('M') df['Sales'] =
df['Quantity'] * df['Price']

# Find the month with the highest total sales best_month
= df.groupby('Month')['Sales'].sum().idxmax()
highest_sales = df.groupby('Month')['Sales'].sum().max()
print(f"Best month: {best_month}")
print(f"Total sales: ${highest_sales:.2f}")

```



4.2.2. Best Selling Product 02:08 and

performs the following operations:

Write a Python program that takes the file name of a CSV file as input, reads the data,

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles
2025-01-04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

```
import pandas as pd
```

```
# Prompt the user for the file name
```

```
file_name = input()
```

```
# Load the data df =
```

```
pd.read_csv(file_name)
```

```
# Find the product with the highest total quantity sold
best_product = df.groupby('Product')['Quantity'].sum().idxmax()
highest_quantity = df.groupby('Product')['Quantity'].sum().max()
# Display the result print(f"Best selling
product: {best_product}") print(f"Total quantity
sold: {highest_quantity}")
```



4.2.3. City that Sold the Most Products

01:45 and performs the following

operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

Write a Python program that takes the file name of a CSV file as input, reads the data,

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los Angeles

2025-01-05,Product A,3,20,Chicago 2025-01-

05,Product C,10,30,Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

```
import pandas as pd
```

```
# Prompt the user for the file name
```

```
file_name = input()
```

```
# Load the data df = pd.read_csv(file_name) #
```

```
Group by City and calculate total quantity
```

```
city_sales = df.groupby('City')['Quantity'].sum()
```

```
# Find the city with the highest total quantity sold
```

```
best_city = city_sales.idxmax()
```

```
# Display the result print(f"City sold the most
```

```
products: {best_city}")
```



4.2.4. Most Frequently Sold Product Pairs

01:25

and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

Sample Data:

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los Angeles

Write a Python program that takes the file name of a CSV file as input, reads the data,

2025-01-05,Product A,3,20,Chicago 2025-01-

05,Product C,10,30,Los Angeles

Explanation:

Transactions:

- 2025-01-01: Product A, Product B
- 2025-01-02: Product A, Product C
- 2025-01-03: Product B, Product A
- 2025-01-04: Product C, Product B 2025-01-05: Product A, Product C

Now, let's count how often the pairs of products appear together:

- Product A and Product B : Appear in transactions on 2025-01-01 and 2025-01-03.
- Product A and Product C : Appear in transactions on 2025-01-02 and 2025-01-05.
- Product B and Product C : Appears in transactions on 2025-01-04.

Most Frequent Product Combinations:

- Product A and Product B (2 times)
- Product A and Product C (2 times)

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

```
import pandas as pd
```

```
from itertools import combinations
```

```
from collections import Counter
```

```
# Prompt user to input the file name
```

```
file_name = input()
```

```
# Read data from the specified CSV file
```

```
df = pd.read_csv(file_name) # write the
```

```
code pair_counter = Counter()
```

```
# Group by Date and generate product pairs
```

```
for _, group in df.groupby('Date'):  
    products = group['Product'].tolist()    pairs =  
    combinations(sorted(products), 2)  
    pair_counter.update(pairs)
```

```
# Find maximum frequency max_count
```

```
= max(pair_counter.values())
```

```
# Output the most frequent product pairs
```

```
for pair, count in pair_counter.items():
```

```
    if count == max_count:
```

```
        print(f'{pair[0]} and {pair[1]}: {count} times')
```

The screenshot displays a test runner interface with the following components:

- Performance Metrics:**
 - Average time: 0.020 s (20.33 ms)
 - Maximum time: 0.039 s (39.00 ms)
- Test Results Summary:**
 - 1 out of 1 shown test case(s) passed
 - 2 out of 2 hidden test case(s) passed
- Test Case Details:**
 - Test case 1** (39 ms) with a **Debug** button.
 - Expected output:**
 - sales_data.csv
 - Product A and Product B: 2 times
 - Product A and Product C: 2 times
 - Actual output:**
 - sales_data.csv
 - Product A and Product B: 2 times
 - Product A and Product C: 2 times
- Navigation:** Tabs for **Terminal** and **Test cases** are visible at the bottom.

4.2.5. Titanic Dataset Analysis and Data Cleaning

01:36

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using `.info()`).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the 'Age' column with the median age.
8. Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
9. Drop the 'Cabin' column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
 1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
 2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
 17599,71.2833,C85,C
 3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
 4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
 5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
 6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
 7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
 8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
 9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
 10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd
import numpy as np
```

```
# Load the Titanic dataset data =
pd.read_csv('Titanic-Dataset.csv') # 1.
Display the first 5 rows of the dataset
print(data.head())
```

```
# 2. Display the last 5 rows of the dataset
print(data.tail())
```

```
# 3. Get the shape of the dataset
print(data.shape)
```

```
# 4. Get a summary of the dataset (info)
print(data.info())
```

5. Get basic statistics of the dataset

```
print(data.describe())
```

6. Check for missing values

```
print(data.isnull().sum())
```

7. Fill missing values in the 'Age' column with the median age

```
data['Age'] = data['Age'].fillna(data['Age'].median())
```

8. Fill missing values in the 'Embarked' column with the mode

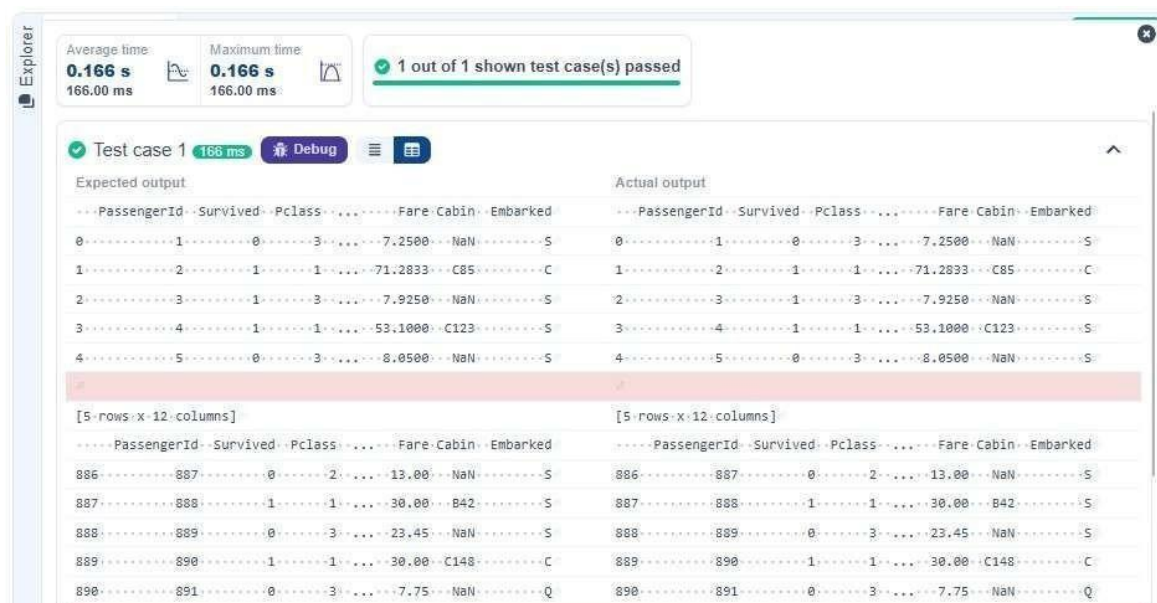
```
data['Embarked'] = data['Embarked'].fillna(data['Embarked'].mode()[0])
```

9. Drop the 'Cabin' column due to many missing values

```
data = data.drop(columns=['Cabin'])
```

10. Create a new column 'FamilySize' by adding 'SibSp' and 'Parch'

```
data['FamilySize'] = data['SibSp'] + data['Parch']
```



The screenshot shows a Jupyter Notebook interface with a test case summary at the top indicating '1 out of 1 shown test case(s) passed'. Below this, a table compares 'Expected output' and 'Actual output' for a dataset with 5 rows and 12 columns. The columns are PassengerId, Survived, Pclass, Fare, Cabin, and Embarked. The data is consistent between the expected and actual outputs, with the only difference being the 'Cabin' column, which contains 'NaN' values in the expected output and is missing in the actual output.

PassengerId	Survived	Pclass	Fare	Cabin	Embarked
0	1	0	7.2500	NaN	S
1	2	1	71.2833	C85	C
2	3	1	7.9250	NaN	S
3	4	1	53.1000	C123	S
4	5	0	8.0500	NaN	S

PassengerId	Survived	Pclass	Fare	Cabin	Embarked
886	887	0	13.00	NaN	S
887	888	1	30.00	B42	S
888	889	0	23.45	NaN	S
889	890	1	30.00	C148	C
890	891	0	7.75	NaN	Q

Average time
0.166 s
166.00 ms



Maximum time
0.166 s
166.00 ms



✓ 1 out of 1 shown test case(s) passed

```
[5 rows x 12 columns]                                [5 rows x 12 columns]
(891, 12)                                              (891, 12)
<class 'pandas.core.frame.DataFrame'>                <class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890                    RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):                      Data columns (total 12 columns):
#   Column  ... Non-Null Count  Dtype  ...   #   Column  ... Non-Null Count  Dtype  ...
...   ...   ...              ...   ...   ...   ...   ...              ...   ...
0   PassengerId  891 non-null  int64  ...   0   PassengerId  891 non-null  int64  ...
1   Survived    891 non-null  int64  ...   1   Survived    891 non-null  int64  ...
2   Pclass      891 non-null  int64  ...   2   Pclass      891 non-null  int64  ...
3   Name        891 non-null  object ...   3   Name        891 non-null  object ...
4   Sex         891 non-null  object ...   4   Sex         891 non-null  object ...
5   Age         714 non-null  float64 ...   5   Age         714 non-null  float64 ...
6   SibSp       891 non-null  int64  ...   6   SibSp       891 non-null  int64  ...
7   Parch       891 non-null  int64  ...   7   Parch       891 non-null  int64  ...
8   Ticket      891 non-null  object ...   8   Ticket      891 non-null  object ...
9   Fare        891 non-null  float64 ...   9   Fare        891 non-null  float64 ...
10  Cabin       204 non-null  object ...   10  Cabin       204 non-null  object ...
11  Embarked    889 non-null  object ...   11  Embarked    889 non-null  object ...
dtypes: float64(2), int64(5), object(5)              dtypes: float64(2), int64(5), object(5)
memory usage: 66.2+ KB                               memory usage: 66.2+ KB
None                                                  None
... PassengerId ... Survived ... Pclass ... SibSp ... P
arch ... Fare
count 891.000000 891.000000 891.000000 ... 891.000000 891.00
0000 891.000000
mean 446.000000 0.383838 2.308642 ... 0.523008 0.38
1594 32.204208
std 257.353842 0.486592 0.836071 ... 1.102743 0.80
6057 49.693429
count 891.000000 891.000000 891.000000 ... 891.000000 891.00
0000 891.000000
mean 446.000000 0.383838 2.308642 ... 0.523008 0.38
1594 32.204208
std 257.353842 0.486592 0.836071 ... 1.102743 0.80
6057 49.693429
```

Explorer

Average time
0.166 s
166.00 ms

Maximum time
0.166 s
166.00 ms

1 out of 1 shown test case(s) passed

Debugger

count	891.000000	891.000000	891.000000	...	891.000000	891.000000	count	891.000000	891.000000	891.000000	...	891.000000	891.000000
mean	446.000000	0.383838	2.308642	...	0.523008	0.381594	mean	446.000000	0.383838	2.308642	...	0.523008	0.381594
std	257.353842	0.486592	0.836071	...	1.102743	0.806057	std	257.353842	0.486592	0.836071	...	1.102743	0.806057
min	1.000000	0.000000	1.000000	...	0.000000	0.000000	min	1.000000	0.000000	1.000000	...	0.000000	0.000000
25%	223.500000	0.000000	2.000000	...	0.000000	0.000000	25%	223.500000	0.000000	2.000000	...	0.000000	0.000000
50%	446.000000	0.000000	3.000000	...	0.000000	0.000000	50%	446.000000	0.000000	3.000000	...	0.000000	0.000000
75%	668.500000	1.000000	3.000000	...	1.000000	0.000000	75%	668.500000	1.000000	3.000000	...	1.000000	0.000000
max	891.000000	1.000000	3.000000	...	8.000000	6.000000	max	891.000000	1.000000	3.000000	...	8.000000	6.000000

[8 rows x 7 columns]

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2
dtype	int64

Terminal

Test cases

< Prev

Reset

Submit

Next >

9:37 PM

4/24/2026

4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

02:18

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
2. Convert the 'Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the 'Embarked' column, dropping the first category.
4. Get the mean age of passengers.
5. Get the median fare of passengers.
6. Get the number of passengers by class.
7. Get the number of passengers by gender.
8. Get the number of passengers by survival status.
9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
 4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
 5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
 6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
 7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
 8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
 9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
 10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd
```

```
import numpy as np
```

```
# Load the Titanic dataset data = pd.read_csv('Titanic-
Dataset.csv') data['FamilySize'] = data['SibSp'] +
data['Parch'] # 1. Create a new column 'IsAlone' (1 if
alone, 0 otherwise) data['IsAlone'] =
np.where(data['FamilySize'] == 0, 1, 0)
```

```
# 2. Convert 'Sex' to numeric (male: 0, female: 1)
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
```

```
# 3. One-hot encode the 'Embarked' column data =
pd.get_dummies(data, columns=['Embarked'], drop_first=True)
```

```
# 4. Get the mean age of passengers
print(data['Age'].mean())
```

```
# 5. Get the median fare of passengers
print(data['Fare'].median())
```

6. Get the number of passengers by class

```
print(data['Pclass'].value_counts())
```

7. Get the number of passengers by gender

```
print(data['Sex'].value_counts())
```

8. Get the number of passengers by survival status

```
print(data['Survived'].value_counts())
```

9. Calculate the survival rate

```
print(data['Survived'].mean())
```

10. Calculate the survival rate by gender

```
print(data.groupby('Sex')['Survived'].mean())
```

Average time: 0.079 s, Maximum time: 0.079 s, 1 out of 1 shown test case(s) passed

Test case 1, 79 ms, Debug

Expected output	Actual output
29.69911764705882	29.69911764705882
14.4542	14.4542
3...491	3...491
1...216	1...216
2...184	2...184
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64
0...577	0...577
1...314	1...314
Name: Sex, dtype: int64	Name: Sex, dtype: int64
0...549	0...549
1...342	1...342
Name: Survived, dtype: int64	Name: Survived, dtype: int64
0.3638383838383838	0.3638383838383838
Sex	Sex
0...0.168908	0...0.168908
1...0.742038	1...0.742038
Name: Survived, dtype: float64	Name: Survived, dtype: float64

4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

03:10

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina

Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd
```

```
import numpy as np
```

```
# Load the Titanic dataset data = pd.read_csv('Titanic-Dataset.csv')
```

```
data['FamilySize'] = data['SibSp'] + data['Parch'] data['IsAlone'] =
```

```
np.where(data['FamilySize'] > 0, 0, 1) data = pd.get_dummies(data,  
columns=['Embarked'], drop_first=True)
```

```
# 1. Calculate the survival rate by class
```

```
print(data.groupby('Pclass')['Survived'].mean())
```

```
# 2. Calculate the survival rate by embarked location
```

```
print(data.groupby('Embarked_S')['Survived'].mean())
```

```
# 3. Calculate the survival rate by family size
```

```
print(data.groupby('FamilySize')['Survived'].mean())
```

```
# 4. Calculate the survival rate by being alone
```

```
print(data.groupby('IsAlone')['Survived'].mean())
```

```
# 5. Get the average fare by class
```

```
print(data.groupby('Pclass')['Fare'].mean())
```

```
# 6. Get the average age by class
```

```
print(data.groupby('Pclass')['Age'].mean())
```

```
# 7. Get the average age by survival status
```

```
print(data.groupby('Survived')['Age'].mean()) # 8. Get the average fare by survival
```

```
status print(data.groupby('Survived')['Fare'].mean())
```

9. Get the number of survivors by class

```
print(data[data['Survived'] == 1]['Pclass'].value_counts())
```

10. Get the number of non-survivors by class

```
print(data[data['Survived'] == 0]['Pclass'].value_counts())
```

The screenshot displays a Jupyter Notebook interface with a test runner at the top. It shows 'Average time: 0.103 s' and 'Maximum time: 0.103 s', with a status '1 out of 1 shown test case(s) passed'. Below this, 'Test case 1' is expanded, comparing 'Expected output' and 'Actual output' for a pandas DataFrame. The DataFrame contains columns: Pclass, Survived, Embarked, FamilySize, IsAlone, Fare, and Pclass. The data is identical in both outputs, indicating a successful test case.

Expected output	Actual output
Pclass	Pclass
1 --- 0.629630	1 --- 0.629630
2 --- 0.472826	2 --- 0.472826
3 --- 0.242363	3 --- 0.242363
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Embarked_S	Embarked_S
0 --- 0.506073	0 --- 0.506073
1 --- 0.336957	1 --- 0.336957
Name: Survived, dtype: float64	Name: Survived, dtype: float64
FamilySize	FamilySize
0 --- 0.303538	0 --- 0.303538
1 --- 0.552795	1 --- 0.552795
2 --- 0.578431	2 --- 0.578431
3 --- 0.724138	3 --- 0.724138
4 --- 0.200000	4 --- 0.200000
5 --- 0.136364	5 --- 0.136364
6 --- 0.333333	6 --- 0.333333
7 --- 0.000000	7 --- 0.000000
10 --- 0.000000	10 --- 0.000000
Name: Survived, dtype: float64	Name: Survived, dtype: float64
IsAlone	IsAlone
0 --- 0.505650	0 --- 0.505650
1 --- 0.303538	1 --- 0.303538
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Pclass	Pclass
1 --- 84.154687	1 --- 84.154687
2 --- 20.662183	2 --- 20.662183
3 --- 13.675550	3 --- 13.675550
Name: Fare, dtype: float64	Name: Fare, dtype: float64
Pclass	Pclass
1 --- 38.233441	1 --- 38.233441
2 --- 29.877630	2 --- 29.877630

Explorer

Average time
0.103 s
103.00 ms

Maximum time
0.103 s
103.00 ms

1 out of 1 shown test case(s) passed

Debugger

5 ... 0.136364	5 ... 0.136364
6 ... 0.333333	6 ... 0.333333
7 ... 0.000000	7 ... 0.000000
10 ... 0.000000	10 ... 0.000000
Name: Survived, dtype: float64	Name: Survived, dtype: float64
IsAlone	IsAlone
0 ... 0.505650	0 ... 0.505650
1 ... 0.303538	1 ... 0.303538
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Pclass	Pclass
1 ... 84.154587	1 ... 84.154587
2 ... 20.662183	2 ... 20.662183
3 ... 13.675550	3 ... 13.675550
Name: Fare, dtype: float64	Name: Fare, dtype: float64
Pclass	Pclass
1 ... 38.233441	1 ... 38.233441
2 ... 29.877630	2 ... 29.877630
3 ... 25.140620	3 ... 25.140620
Name: Age, dtype: float64	Name: Age, dtype: float64
Survived	Survived
0 ... 30.626179	0 ... 30.626179
1 ... 28.343690	1 ... 28.343690
Name: Age, dtype: float64	Name: Age, dtype: float64
Survived	Survived
0 ... 22.117887	0 ... 22.117887
1 ... 48.395408	1 ... 48.395408
Name: Fare, dtype: float64	Name: Fare, dtype: float64
1 ... 136	1 ... 136
3 ... 119	3 ... 119
2 ... 87	2 ... 87
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64
3 ... 372	3 ... 372
2 ... 97	2 ... 97
1 ... 80	1 ... 80

< Prev

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